

Probing the Geometry of Truth: Consistency and Generalization of Truth Directions in LLMs Across Logical Transformations and Question Answering Tasks

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Abstract

Large language models (LLMs) are trained on extensive datasets that encapsulate substantial world knowledge. However, their outputs often include confidently stated inaccuracies. Earlier works suggest that LLMs encode truthfulness as a distinct linear feature, termed the “truth direction”, which can classify truthfulness reliably. We address several open questions about the truth direction: (i) whether LLMs universally exhibit consistent truth directions; (ii) whether sophisticated probing techniques are necessary to identify truth directions; and (iii) how the truth direction generalizes across diverse contexts. Our findings reveal that not all LLMs exhibit consistent truth directions, with stronger representations observed in more capable models, particularly in the context of logical negation. Additionally, we demonstrate that truthfulness probes trained on declarative atomic statements can generalize effectively to logical transformations, question-answering tasks, in-context learning, and external knowledge sources. Finally, we explore the practical application of truthfulness probes in selective question-answering, illustrating their potential to improve user trust in LLM outputs. These results advance our understanding of truth directions and provide new insights into the internal representations of LLM beliefs.¹

in applications where accuracy is paramount. A growing body of work (Burns et al., 2022; Azaria and Mitchell, 2023; Marks and Tegmark, 2023; Mallen and Belrose, 2023; Bürger et al., 2024) aims to elicit accurate information from LLMs despite untruthful outputs. These studies use lightweight classifiers, often referred to as *probes*, to analyze patterns in the model’s internal representation that reliably indicate truthfulness. Specifically, given a model and a piece of text, an ideal truthfulness probe is able to tell whether the model believes the text conveys truthful content. By achieving empirical success with linear probes, these works generally believe that truthfulness is internally represented as a salient linear feature and manifests as a “truth direction”.

The goal of this work is to conduct a more in-depth study of the truth direction as an inherent property of LLMs. Although previous works undoubtedly help us understand truth directions, they fail to answer the following questions: **(RQ1)** Do LLMs universally represent truthfulness as a linear feature? **(RQ2)** Are simple probing techniques sufficiently expressive to identify truth directions? **(RQ3)** If and when a “truth direction” exists, in what ways does it generalize? We challenge conclusions from prior works based on empirical evidence, provide preliminary answers to the questions above and present novel observations.

In response to **RQ1**, we find that not all LLMs exhibit a consistent “truth direction”, and that this property is closely related to a model’s capability. While prior works often assume the universal existence of truth directions, we challenge this assumption. Our evidence suggests that truthfulness is more consistently represented across logical negations in more capable LLMs. Based on this finding, we question the conclusion of Levinstein and Herrmann (2024), which attributes the generalization failure to limitations in previous probing techniques. Instead, we argue that the inconsis-

1 Introduction

Large language models (LLMs) possess extensive knowledge, as they are trained on immense corpora that encompass a significant portion of world knowledge. However their outputs are not always reliable and are prone to confidently presenting falsehoods (Bender et al., 2021; Evans et al., 2021; Lin et al., 2022; Liu et al., 2023). This unreliability raises critical concerns about the use of LLMs

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¹Our code is public at https://github.com/colored-dye/truthfulness_probe_generalization

tendency lies within the LLM itself and that in answer to **RQ2**, simple supervised probes are sufficiently expressive to identify the truth direction when it is distinctly represented within the model.

In addressing **RQ3**, we aim to explore the generalization capability of the truth direction, as it illuminates whether an LLM consistently represents truthfulness across different knowledge domains, logical transformations, syntax forms and grounding knowledge source. Previous studies on truth directions have extensively explored the former two aspects of generalization. Regarding syntax forms, they focus either on declarative statements or on (Q, A) pairs; we bridge this gap by testing whether truthfulness probes trained on simple statements generalize to question answering (QA) tasks. Additionally, we examine several variations of QA, including zero- and few-shot QA, with and without provided answer options, and grounded either in parametric knowledge or in question contexts. Our experimental results show that truthfulness probes demonstrate a high degree of generalization.

Furthermore, based on our observation that truthfulness probes are calibrated on certain QA tasks, we introduce their use in selective QA. In this application, we select the subset of answers evaluated as correct by the truthfulness probe from those generated by the LLM. Through this demonstration, we aim to show how truthfulness probes can enhance user trust in real-world LLM-based applications.

Our contributions are as follows:

1. We summarize truthfulness probes from prior works, introduce a new instantiation, and conduct extensive experiments.
2. We study truth directions following three research questions. In addressing **RQ1**, we explore whether the truth direction is common among LLMs. For **RQ2**, we test if sophisticated probing techniques are required to identify truth directions. In answer to **RQ3**, we assess the generalization capabilities of truth directions.
3. We demonstrate a practical application of truthfulness probes for selective question answering, improving generation quality by filtering out unreliable answers.

2 Related Work

Eliciting latent knowledge (ELK). The field of scalable oversight (Christiano et al., 2021) seeks to address the information asymmetry between su-

perhuman AI systems and human evaluators. It is assumed that although the AI possesses significant knowledge, its behavior is untrustworthy because it is not trained with an objective that explicitly incentivizes outputs to align with the truth (Mallen and Belrose, 2023). ELK is an approach within scalable oversight that aims to identify patterns in an AI’s activations that correspond to the truth. The primary challenge lies in identifying patterns that generalize reliably to questions where human evaluators are unable to verify the answers (Mallen and Belrose, 2023). Our work demonstrates that probing techniques can achieve reasonable generalization with limited supervision, suggesting that probes may offer a promising approach for ELK.

Probing for truthfulness. Several studies have used probing techniques to uncover truthfulness by examining an LLM’s internal states, independent of its inputs or outputs (Lee et al., 2023; Joshi et al., 2024). Regarding the *geometry of the representation* of truthfulness, most studies agree that this representation is likely linear, as demonstrated by the use of linear probes in studies such as CCS (Burns et al., 2022), mass-mean (Marks and Tegmark, 2023; Li et al., 2023), TTPD (Bürger et al., 2024) and the commonly used baseline: logistic regression. Notably, CCS is an unsupervised approach and targets yes/no *question-answering tasks*, while others are supervised and target *factual statements*. In contrast, some works are *geometry-agnostic*. Azaria and Mitchell (2023) propose SAPLMA which is based on MLP architecture, while He et al. (2024) introduce the LLM Factoscope, which leverages a Convolutional Neural Network architecture.

The aforementioned studies examine truthfulness grounded in a model’s *parametric knowledge*, whereas Sky et al. (2024) detect hallucination with probes in the setting of in-context generation, where *knowledge is grounded in the context*.

3 Summary of Probes and Data

In this section, we formally define the task of probing binary features from a model’s internal representations and describe the specific probe architectures used in this study. Furthermore, we detail the labeled datasets used to train and evaluate the truthfulness probes.

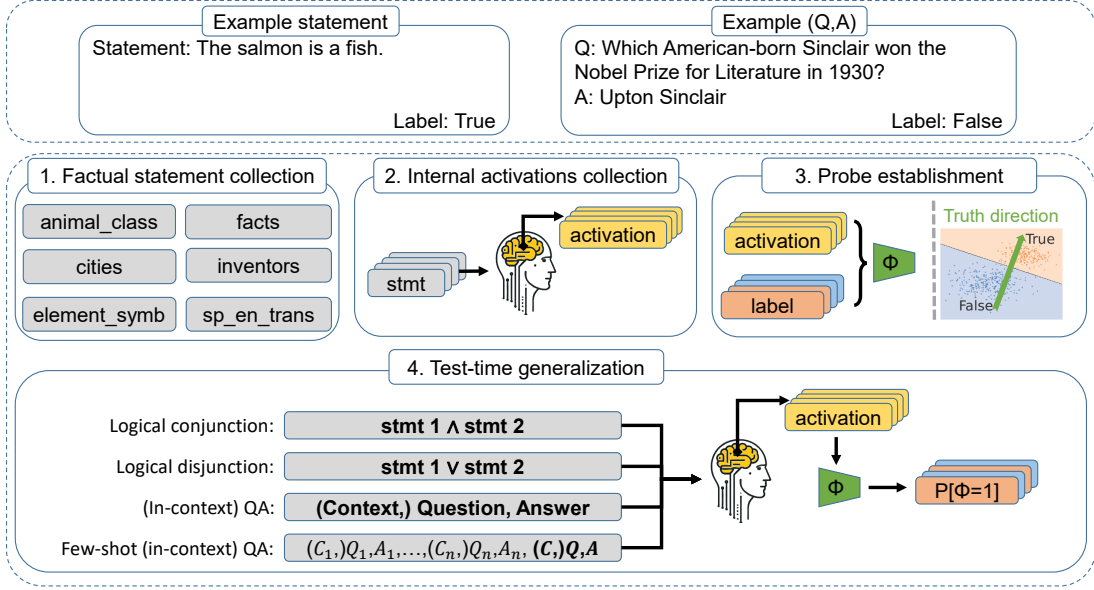


Figure 1: Illustration of truthfulness probes. A truthfulness probe is established using the LLM’s internal states when processing labeled statements. The probe is then able to tell if the LLM believes an unseen statement or a given response to a question is true or false leveraging only the LLM’s internal states.

3.1 Formulation of Binary Probes

Our target model is the transformer language model (Vaswani, 2017), which processes token sequences through a series of layers. A sequence of n input tokens, $t = (t_1, t_2, \dots, t_n)$, is first converted into embeddings, $\mathbf{h}^{(0)} = (\mathbf{h}_1^{(0)}, \mathbf{h}_2^{(0)}, \dots, \mathbf{h}_n^{(0)})$, by the initial embedding layer. The embeddings are then passed through L layers, where each layer generates representations based on the preceding layer’s output. The representation of a single token is a vector: $\mathbf{h}_i^{(j)} \in \mathbb{R}^d$ and $(0 \leq j \leq L, 1 \leq i \leq n)$. Finally, the LLM produces predictions using $\mathbf{h}^{(L)}$.

Suppose we have prior knowledge that the model internally represents a binary feature. Our goal is to establish a probe Φ , that uses only the model’s representations to classify the target attribute. The probe outputs either binary labels $-1/1$, or probabilistic predictions such as $P[\Phi = 1]$.

For autoregressive models, which are the primary focus of this paper, we utilize the representation at the final token position of the l -th layer, $\mathbf{h}_{-1}^{(l)}$. This approach aligns with prior work (Burns et al., 2022; Azaria and Mitchell, 2023; Marks and Tegmark, 2023), where the final token position attends to all previous tokens due to the causal attention mechanism. We also assume we have a labeled dataset, $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^M$, where x_i represents a token sequence and y_i is the label for the target attribute. Processing these sequences through the model yields $\mathcal{D}_{\text{rep}} = \{((\mathbf{h}_{-1}^{(l)})_i, y_i)\}_{i=1}^M$, which we

use to build our classifier.

To maximize classification accuracy, we define a cost function to quantify the classification error of the probe, $J(\Phi, \mathbf{h}, y)$. The mechanistic objective is to minimize the expected classification error over the data distribution: $\arg \min_{\Phi} \frac{1}{M} \sum_{i=1}^M J(\Phi, \mathbf{h}_i, y_i)$.

3.2 Instantiations of Binary Probes

We classify the probes into two categories: geometry-oriented and statistics-based. Geometry-oriented probes leverage knowledge of the geometric structure of the representation. Under the “truth direction” hypothesis, true/false representations can be separated by a hyperplane, and the normal vector of this hyperplane corresponds to the “truth direction”. In contrast, statistics-based probes are geometry-agnostic and aim to maximize the probability of observing the correct labels given the input data. Our implementations are based on the scikit-learn (Pedregosa et al., 2011) library.

Geometry-oriented Probes. For geometry-oriented probes, we introduce two instantiations: linear support vector machine (SVM) (Cortes and Vapnik, 1995) and mass-mean (MM) (Marks and Tegmark, 2023) instantiation. The rationale for selecting linear SVM is its ability to maximize the margin, which aligns with the goal of identifying a separating hyperplane. As SVM does not directly provide probability predictions, we fit a post-hoc